Bearing Capacity Check (Terzaghi + EC7 DA1)

This notebook performs shallow foundation bearing checks using the Terzaghi general equation with EC7 Design Approach 1 combinations:

- `DA1-1 = A1 + M1 + R1`
- `DA1-2 = A2 + M2 + R1`

Assumptions in this template:

- Vertical centric load only (`i` factors = 1.0)
- Drained parameters (`c'` , `phi'`)
- Effective stress method
- Rectangular footing with Terzaghi-style shape/depth factors

```
Calc environment ready.
```

Inputs

Use SI base units: `kN` , `m` , `kN/m^3` , `kPa` .

DA1-1 (`A1 + M1 + R1`)

$$
\begin{aligned}
V_{Ed} &= 1665.000 \\
A &= B \cdot L = 2.000 \cdot 3.000 && = 6.000 \\
q_{Ed} &= \frac{V_{Ed}}{A} = \frac{1665.000}{6.000} && = 277.500 \\
c_d &= \frac{c_k}{\mathrm{M1}} = \frac{5.000}{'gamma_{c':1.0,'gamma_{phi':1.0}} && = 5.000 \\
\mathrm{tanphi}_d &= \frac{\tan\left(\mathrm{radians}\ \phi_{k_{deg}}\right)}{\mathrm{M1}} = \frac{\tan(\mathrm{radians}\ 32.000)}{'gamma_{c':1.0,'gamma_{phi':1.0}} && = 0.625 \\
\phi_d &= \mathrm{degrees}(\arctan \mathrm{tanphi}_d) = \mathrm{degrees}(\arctan 0.625) && = 32.000 \\
q_{ult} &= 1065.658 \\
q_{Rd} &= \frac{q_{ult}}{\gamma_{Rv}} = \frac{1065.658}{1.000} && = 1065.658 \\
\mathrm{util} &= \frac{q_{Ed}}{q_{Rd}} = \frac{277.500}{1065.658} && = 0.260
\end{aligned}
$$

DA1-2 (`A2 + M2 + R1`)

$$
\begin{aligned}
V_{Ed} &= 1290.000 \\
A &= B \cdot L = 2.000 \cdot 3.000 && = 6.000 \\
q_{Ed} &= \frac{V_{Ed}}{A} = \frac{1290.000}{6.000} && = 215.000 \\
c_d &= \frac{c_k}{\mathrm{M2}} = \frac{5.000}{'gamma_{c':1.25,'gamma_{phi':1.25}} && = 4.000 \\
\mathrm{tanphi}_d &= \frac{\tan\left(\mathrm{radians}\,\phi_{k_{deg}}\right)}{\mathrm{M2}} = \frac{\tan(\mathrm{radians}\,32.000)}{'gamma_{c':1.25,'gamma_{phi':1.25}} && = 0.500 \\
\phi_d &= \mathrm{degrees}(\arctan \mathrm{tanphi}_d) = \mathrm{degrees}(\arctan 0.500) && = 26.560 \\
q_{ult} &= 539.801 \\
q_{Rd} &= \frac{q_{ult}}{\gamma_{Rv}} = \frac{539.801}{1.000} && = 539.801 \\
\mathrm{util} &= \frac{q_{Ed}}{q_{Rd}} = \frac{215.000}{539.801} && = 0.398
\end{aligned}
$$

Summary Table

	Case	phi_d_deg	c_d_kPa	V_Ed_kN	q_Ed_kPa	q_Rd_kPa	utilization	PASS
0	DA1-1 (A1+M1+R1)	32.00000	5.0	1665.0	277.5	1065.657592	0.260403	True
1	DA1-2 (A2+M2+R1)	26.56026	4.0	1290.0	215.0	539.801252	0.398295	True

Notes

- If your National Annex uses different partial factors, edit `A1`, `A2`, `M1`, and `M2` in the input cell.
- This notebook checks bearing resistance only. Settlement and sliding/overturning must be checked separately.
- For eccentric or inclined loading, add inclination/eccentricity factors before finalizing design resistance.